



Pranay Tummalapalli

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ABOUT ME

Robotics engineer with five years building systems that have to work outside the lab: surgical robots taken through clinical validation, autonomous driving in Indian traffic, and LLM inference benchmarked on edge hardware. Focused on rigorously evaluating how vision-language-action models hold up under real-world perturbation, particularly in precision-critical domains like surgical manipulation.

EDUCATION AND TRAINING

01/08/2017 – 01/07/2021 New Delhi, India

BACHELOR OF TECHNOLOGY Bharati Vidyapeeth's College of Engineering

Website <https://bvcoend.ac.in/> | **Field of study** Electronics and Communications | **Final grade** 7.36 | **Level in EQF** EQF level 6 |

National classification 6 | **Type of credits** CBCS | **Number of credits** 216 |

Thesis Biomimetic arm to convert speech to sign language

14/09/2026 – CURRENT Bristol, United Kingdom

MASTERS OF SCIENCE University of Bristol

Website <https://www.bristol.ac.uk/study/postgraduate/taught/msc-robotics/> | **Field of study** Robotics | **Level in EQF** EQF level 7

LANGUAGE SKILLS

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

WORK EXPERIENCE

AI RESEARCH CONSULTANT – CONSCIOUS ENGINES – 15/01/2026 – 30/04/2026 – BENGALURU, INDIA

- Led a benchmarking study of on-device LLM inference for always-on personal agents; first author on the resulting paper (arXiv, 2026): <https://arxiv.org/abs/2603.23640>
- Deployed Qwen 2.5 1.5B (4-bit quantised) on four edge platforms: Raspberry Pi 5 with Hailo-10H NPU, Samsung Galaxy S24 Ultra, iPhone 16 Pro and NVIDIA RTX 4050 laptop GPU.
- Designed a repeatable benchmark: fixed 258-token prompt, 20 warm runs per device, measuring throughput, latency, power and thermal behaviour.

Analysed the performance limits of each platform:

- Mobile: thermal throttling limits performance before compute does; the iPhone 16 Pro lost about 41% of its throughput as it heated up.
- Galaxy S24 Ultra: an OS-enforced GPU frequency floor stopped inference entirely.
- RTX 4050: capped by its battery power limit; sustained 131.7 tok/s at 34.1 W.

- Hailo-10H: limited by memory bandwidth; sustained 6.9 tok/s at under 2 W with almost no run-to-run variation.
- Showed that the Hailo-10H matches the RTX 4050 on energy per token despite 19× lower throughput.

SR. EMBEDDED FIRMWARE ENGINEER, VEHICLE CONTROL – MINUS ZERO ROBOTICS PVT. LTD. – 04/12/2024 – 30/04/2026 – BENGALURU, INDIA

Business or Sector: Other service activities | **Department:** Engineering | **Email:** pranay@minuszero.ai

- Design and development of a vehicle control module that runs on a jetson Orin AGX platform, as part of the onboard software stack for autonomous vehicles. Tested on public roads on Tata Nexon.
- CAN/UDP socket programming to interface onboard software stack with Tata Nexon Drive-by-wire system.
- Vehicle module, onboard software inspired from Baidu's Apollo project architecture for autonomous Driving.
- Designed and tested vehicle motion control and path planning algorithms like pure-pursuit for lateral control and PID control for longitudinal control.
- Thorough understanding of system design for autonomous vehicle motion control.
- Publisher-subscriber architecture and logging using eCAL framework and message framework using protobuf 3.0.
- Experience developing modular and scalable software.
- Heading the Design and development of in-house STM32G4 based drive-by-wire system for Tata Nexon:
 - Since tata nexon is not an ADAS-compatible vehicle, it does not have a drive-by-wire system
 - CAN sniffing using SavvyCan and Peak-Can for vehicle reports and module feedbacks
 - Interface with Steering EPAS and steering encoder to design a closed loop PID control.
 - Interface with throttle ECU to send throttle percentage values using DACs on STM32.
 - Brake actuated using an external high-speed linear actuator to pull the brake pedal through a capstan mechanism.
 - State machine architecture to ensure SOP, fault handling, and ensuring safety mechanisms.
- Headed technical collaborations with different OEMs and vendors:
 - Mercedes Benz Research and Development Institute for L3 ADAS for Indian roads on S-Class.
 - Ashok Leyland for autonomous transport for ports.
 - ARAI for drive-by-wire system for Tata Nexon

SR. EMBEDDED FIRMWARE ENGINEER, MOTOR CONTROL – MERIL HEALTHCARE (ROBOTIC ENDOSURGERY) – 15/11/2023 – 15/11/2024 – CHENNAI, INDIA

- STM32 based solutions for EtherCAT Slave implementation for ASIX58400 based in-house motor drives, RFID Data reading etc. implemented on the system.
- Motor control code for ethercat based Motor drive from Novanta, and Maxon. Developed multi slave code with Distributed Clocks for Robot end-effector actuators.
- Used C in linux for development. Integrated with shared memory and UDP server-client architecture for sharing desired setpoint data to the motor.
- Mathematical modeling of motor and motor and drive characteristics analysis for PID control.
- Integrating Surgical robotic system for laparoscopic surgery with touchscreen, x86 single-board-computer, and optimizing Ubuntu operating system to run GUI.
- Part selection based on identified system requirements, supplier identification to optimize for cost, lead time and long term support. Found a long term solution with low cost high performance parts.
- Contributed towards the success of the MISSO orthopedic surgical robot and its product launch.
- International Experience: Part of delegation to Shanghai to attend the CMEF 24 and identify potential partners. Converted 2 suppliers to long term partners.
- Part of a team to find Chinese robotics companies including humanoids, quadrupeds, rehabilitation exoskeletons, and autonomous vehicles to partner with for OEM development. Visited 15 companies across Shanghai, Beijing, Shenzhen, Longyan, Suzhou.
- Quality and Regulatory Experience: Formulated a system BOM for BOM based purchase system and system versioning. Formed requirements for each subsystem and implemented test methods to ensure requirements are met.
- Organized Cadaver trials for the laparoscopic and orthopedic robots and implemented on field research for user validation with surgeons.

SYSTEMS ENGINEER – ARTICULUS SURGICAL PVT. LTD – 01/01/2023 – 24/10/2023 – BENGALURU, INDIA

- incharge of day-to-day R&D with a multidisciplinary team, feature pipelining for Patient Cart, Endoscopic vision and Surgeon's console
- Incharge of multidisciplinary design of Surgeon's console including usability, electronics, master-slave forward kinematics using DH Parameters, ergonomics etc.
- PID based control system for BLDC control with magnetic encoder.
- Implemented ESD, EMI/EMC protection in the PCBs
- Requirements engineering for the product and usability design, preliminary design for verification and validation tests, and MDR compliance
- End-user validation with Ob/gyn, Urology and GI surgeons.
- Basic understanding of harmonized MDR standards including IEC 60601, ISO 14971 etc.

MECHATRONICS & INDUSTRIAL DESIGN LEAD – ARTICULUS SURGICAL PVT. LTD – 19/11/2021 – 01/01/2023 – BENGALURU, INDIA

- Part of the founding team and contribution toward early stage development of Robotic Surgery System.
- Developed a prototype for the Surgeon's Console: design and physical analysis of gravity compensation mechanism for a 7-DOF console manipulator; integration with rotary encoders for joint positions; angular monostable positioning of joints
- Platforms and Skills:
 - Autodesk Fusion360 - CAD design for complex multi-body assemblies, joint designs, simulations and FEM, rendering tool.
 - FDM 3D Printing: Prusa I3 Mk3, Creality ender 3 v2, CR10Max. Prusa and Cura slicer.
 - SLA 3D Printing: Elegoo Mars 3 and slicer
 - Industrial Design - work samples on www.articulussurgical.com
- DFM - 3 axis milling, Lathe/turning, SLS for Titanium, FDM Plastic 3D Printing

CO-FOUNDER – AKSHAR BIONICS (FUNDED BY MINISTRY OF EDUCATION) – 01/01/2021 – 31/10/2021 – NEW DELHI, INDIA

- Used Biomimetic design methodology to develop fully 3D-printed and actuated upper-body robotic arm
- Developed Joint mechanisms and calculated inverse kinematics using DH-Parameters.
- Patents for Software integrated system and robotic arm design for the project pending.
- The project won Smart India Hackathon 2018 (Hardware Edition) and subsequently was granted funding from MOE, GOI.

PRODUCT DESIGN INTERN – TRESTLE LABS – 20/02/2020 – 20/07/2020 – NEW DELHI, INDIA

- Designed a phone stand to enable visually impaired users to scan and convert hardcopy documents and books into audio format (audiobooks) using Trestle Labs' Kibo XL app.
- Carried out User research and validation testing at NGOs in New Delhi

● **PROJECTS**

01/09/2025 – CURRENT

Latty, A ROS based tricycle platform

- A front wheel drive non-holonomic tricycle platform to develop and experiment with different path planning algorithms
- urdf/xacro model definitions for Gazebo Classic
- Lower level controllers for joint state publishers and reports that interact with the /controller_manager
- ROS2 node for joint control:
 - Publishers to the joint state topics
 - Subscribers for interfacing with the robot and reading model state
- State estimation and localisation techniques for robot pose estimation:
 - Euler and exponential map integration for yaw estimation from IMU.
 - Linear Kalman Filter and EKF with IMU and wheel odometry.
- Pointcloud data from LIDAR for mapping.
- ReRun for visualisation.

<https://github.com/pranaytummalapalli/latty>

01/01/2021 – 31/10/2021

Vocol, Akshar Bionics

- Vocol is a system that converts speech input to American Sign Language and Indian Sign Language on a Life-Scale humanoid robot designed using biomimetic design methodology for its joints and limb movements. Servo actuated joints and I2C networked servo controllers controlled by an STM microcontroller.

Link <https://youtube.com/shorts/trUFDeqpUHU?feature=share>

15/01/2020 – 10/02/2020

LoRa Sensor Network for Accident Prevention, PriorFire

- A Sensor network between multiple cars that classify a car as a danger if the driver doesn't drive properly, and then informs the other cars from a long distance that there is a dangerous driver nearby, increasing their reaction time.
- Used LoRaWAN protocol, Image Processing for drowsiness detection, lane detection using openCV, and position encoder to calculate steering wheel deviation. NodeMCU, LoRa RA-02, LCD, Raspberry pi 3B
- Won 1st prize in Hardware Productathon, E-Summit 2020, IIT-Roorkee

Smart Object Detection and Guidance Device for Visually Impaired, Raah

- A two stage haptic feedback based algorithm to guide a blind person to an object. It uses object recognition in OpenCV using tensorflow and deep learning.

● HOBBIES AND INTERESTS

Motorcycling

Motorcycling has become one of the most grounding and meaningful parts of my life. Over the last six months, I've travelled close to 7,000 km around Bengaluru — exploring forests, hill stations, tiny villages, and winding backroads. Long-distance touring is incredibly relaxing for me: the steady hum of the engine, the open road, and the changing landscapes create a sense of freedom that's hard to find elsewhere. Riding teaches patience, focus, and calm decision-making, and it has a way of slowing life down in the best possible way. Along the way, I've met countless riders and locals, each with their own stories, reminding me how travel connects people from all walks of life. For me, motorcycling isn't just a hobby — it's a way to reset, reflect, and reconnect with the world around me.

Tabla, Instrumental Music

I have been learning classical tabla since the age of eight under the guidance of my guruji, who belongs to the Farukhabad gharana, one of the six principal gharanas of Hindustani classical music.

My training follows the traditional guru-shishya parampara, where knowledge is passed down orally through close mentorship. Over the years, I have performed at several auditoriums, presenting classical solo tabla pieces and refining my technique. I also secured 3rd place at the zonal-level tabla solo competition in Delhi, a meaningful milestone in my musical journey.

Badminton, Sports

In 2018, as part of the National Service Scheme, I actively contributed to the Swachh Bharat Mission, supporting efforts to promote cleanliness and hygiene in local communities. This experience allowed me to be part of a national initiative and make a meaningful impact through hands-on service.

● HONOURS AND AWARDS

31/07/2021

Best student award for Electronics and Communications Engineering Dept – Bharati Vidyapeeth's College of Engineering, New Delhi

Awarded for excellence in overall achievements during bachelor's degree for academic, co-curricular, contribution to college ecosystem and technical societies

02/02/2020

1st Position, Hardware Productathon, E-Summit 2020, IIT Roorkee – IIT Roorkee

Project PriorFire - LoRaWAN based sensor network for accident prevention

22/06/2018

1st Position, Smart India Hackathon Hardware Edition 2018 (Drones and Robotics Track, IIT Kanpur) – Ministry of Education, Government of India

Project Lingua - Akshar Bionics.

Project funded by Ministry of Education, Govt. Of India

10/03/2018

1st Position, Technical Projects Exhibition, IEEE-MSIT – IEEE-MSIT

Project SSDr - Safety and Surveillance rover for disaster affected zones.

08/03/2018

2nd Position, Technical Projects Exhibition, Fervour X, BVPIEEE – BVPIEEE

Project SSDr - Safety and Surveillance rover for disaster affected zones.